PEEK: One-Step-Look-Ahead Exposure Bias Correction for Diffusion Sampling

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THE GAP

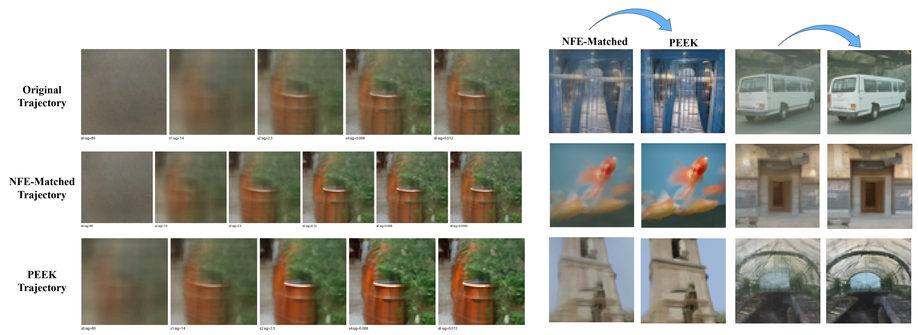
Trained on real-data-noise states. Sampled on model-made states.

In few-step sampling, each large transition depends on earlier imperfect predictions. The denoiser is therefore queried away from the forward-noised data distribution used in training.

- Early errors propagate through the trajectory.
- Outputs become blurry, distorted, or detail-poor.
- More ordinary steps can continue along the same drifted path.

PEEK spends selected evaluations on a cleaner, higher-signal look-ahead query.

PEEK IN ONE LOOK Training-free; deterministic; sampler plugin



Predict the current clean estimate > **step provisionally** > **query once ahead** > **select by a searched gate** and update from the original state.

AT A GLANCE

49.3%
maximum FID reduction versus an NFE-matched baseline

65.3%
fewer denoiser evaluations for a worst-case 9-step schedule search

3.2 min
maximum one-time calibration reported in the experiments

SEARCH WHERE TO PEEK

A guarded, metric-driven schedule

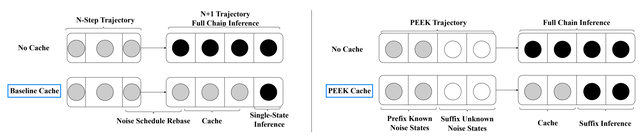
Each gate is tested once from left to right. A look-ahead is accepted only when it beats an ordinary solver using the same denoiser-evaluation budget.

The selected schedule is fixed at deployment; no per-image search is required.

Vanilla PEEK searches binary gates. PEEK-G extends the same idea to solver-direction interpolation with fractional gates.

MAKE CALIBRATION PRACTICAL

Two caches remove repeated prefixes



Baseline rebasing moves cached ordinary states onto the next NFE-matched grid. **Prefix reuse** starts each later PEEK candidate from the last selected state and evaluates only its suffix.

225 to 78 evaluations
worst-case 9-step vanilla search; 65.3% reduction

The baseline rebasing cache is approximate because solver-interval differences can induce small mismatch.

RESULTS

Better use of the same inference budget

Every row compares PEEK with the same solver at matched denoiser evaluations.

Dataset / model	Solver	Matched	PEEK	Gain
ImageNet-EDM 64	DPM-Solver++	19.71	**9.99**	**49.3%**
LSUN-EDM 256	DPM-Solver++	24.17	**13.02**	**46.1%**
LSUN basic 256	DEIS	23.36	**14.18**	**39.3%**
CIFAR-10 32, Model A	DPM-Solver++	25.80	**20.23**	**21.6%**
MS-COCO 512, order 3	DPM-Solver++	13.54	**11.74**	**13.3%**
CelebA-HQ 256	DEIS	23.15	**20.42**	**11.8%**
ImageNet DiT 256	DPM-Solver++	22.65	**21.90**	**3.3%**

FID is lower-is-better. Values and percentage reductions are from the PEEK manuscript.

The signal generalizes

Pixel + latent
32 to 512 resolution

Three settings
unconditional, label, text

Multiple solvers
including higher order

PEEK-G: direction-space extension

PEEK-G blends current and look-ahead solver directions. It lowers FID from **17.50 to 13.16** on CIFAR-10 Model B and from **14.41 to 10.66** on MS-COCO against NFE-matched DPM-Solver++ baselines.

TAKE-HOME

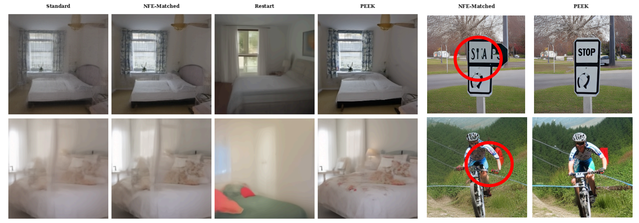
Spend extra inference on a better signal.

PEEK does not merely lengthen an already drifted trajectory. It queries a higher-signal provisional state, injects that generally cleaner estimate back into the current transition, and keeps only positions that beat an ordinary compute-matched step.

TRAINING-FREE • FIXED DEPLOYMENT SCHEDULE • NFE-MATCHED GAINS

QUALITATIVE COMPARISON

Sharper details; more coherent shapes



Paper figure: standard, NFE-matched, Restart-style repeat, and PEEK trajectories. PEEK remains deterministic while using the look-ahead estimate to correct selected transitions.

INTERPRETATION

What changes, and what does not

- **No training or fine-tuning.** PEEK wraps a pretrained deterministic sampler.
- **No new solver derivation.** The current update remains anchored at the original state.
- **A one-time schedule.** Calibration is specific to a model, sampler, step count, and metric.

Limitations. Coverage can decrease, and a confidently biased look-ahead can amplify error.

PROJECT PAGE

See PEEK in motion

Scan for the video, project materials, and updates.
dachengruoque.github.io/PEEK-Video